

TASK 5 : Multi-Factor Authentication (MFA) Comparison Report

1. Multi Factor Authentication.

Multi-Factor Authentication (MFA) is a security method that requires users to verify their identity using two or more different authentication factors before accessing an account, application or a system. MFA provides an additional layer of security beyond a password and helps protect accounts from unauthorized access.

2. Objective

The objective of this report is to understand different authentication factors, compare their advantages and disadvantages, and explain how MFA improves account security.

3. Types of Authentication Factors

Authentication factors are commonly classified into three main categories.

3.1 Knowledge Factor : -Something You Know

This factor uses information that the user knows.

Examples:

- Passwords
- PINs
- Passphrases
- Security codes memorized by the user

Advantages:

- Easy to implement.
- Generally inexpensive.
- Familiar and convenient for users.

Disadvantages:

- Can be guessed or stolen.
- Users may reuse passwords.
- Vulnerable to phishing and social engineering.

3.2 Possession Factor :-Something You Have

This factor uses a physical device or item that the user possesses.

Examples:

- Mobile phones receiving authentication codes
- Authenticator applications
- Hardware security keys
- Smart cards

Advantages:

- Adds a layer of protection beyond passwords.
- Makes account access more difficult for attackers who only know the password.
- Hardware security keys can provide strong resistance to phishing.

Disadvantages:

- Devices can be lost, stolen, or damaged.
- SMS codes may be vulnerable to SIM-swapping attacks.
- Users may experience difficulty if their authentication device is unavailable.

3.3 Inherence Factor :- Something You Are

This factor uses a user's unique biological characteristics.

Examples:

- Fingerprints
- Facial recognition
- Iris recognition

Advantages:

- Convenient and quick to use.
- Does not require memorizing another password.
- Can make unauthorized access more difficult.

Disadvantages:

- Sensors may occasionally fail to recognize the user.
- Biometric information requires careful protection.

- Biometric characteristics cannot be changed as easily as passwords if compromised.

4. Comparison of Authentication Factors

Feature	Knowledge	Possession	Inherence
Based on	Something you know	Something you have	Something you are
Example	Password	Security key	Fingerprint
Main advantage	Simple and familiar	Adds device-based verification	Convenient verification
Main risk	Guessing or theft	Loss or theft of device	Biometric privacy and recognition errors
Security strength	Depends on secrecy and strength	Depends on the method used	Depends on the biometric system

5. How MFA Improves Account Security

MFA improves account security in several ways:

- Additional protection: A stolen password alone may not be enough to access an account.
- Reduced risk of credential theft: Attackers must overcome an additional authentication requirement.
- Protection against password reuse: MFA can help protect accounts even when a reused password is exposed.
- Improved organizational security: MFA helps protect email, banking, cloud services, and administrative accounts.
- Early detection: Unexpected authentication requests may alert users to attempted account access.

6. Comparison of Common MFA Methods

MFA Method	Description	Security Consideration
SMS OTP	A one-time code is sent by text message	Convenient, but vulnerable to SIM swapping and interception
Authenticator app	Generates time-based verification codes	Usually preferable to SMS, but codes can still be phished

Push notification	Sends an approval request to a registered device	Users should reject unexpected requests
Hardware security key	Uses a physical key to verify identity	Strong phishing resistance when properly implemented
Passkey	Uses cryptographic credentials, often unlocked with a PIN or biometric	Phishing-resistant and recommended where supported

7. Is OTP Alone Considered MFA?

A one-time password (OTP) is not automatically MFA.

If a user logs in using only an OTP, it is generally single-factor authentication if that OTP is the only authentication factor used. However, if a user enters a password and then verifies their identity with an independently delivered OTP, the process can qualify as MFA because it combines two authentication factors.

For example:

- Password + authenticator code = two factors.
- Password + security key = two factors.
- Password + fingerprint = two factors.
- OTP alone = one factor, when used by itself.

Two steps do not necessarily mean two factors; the factors must belong to different categories.

8. Best Practices for Using MFA

- Enable MFA on important accounts.
- Prefer passkeys or hardware security keys when available.
- Use authenticator applications rather than SMS where appropriate.
- Never share OTPs or approve unexpected authentication requests.
- Store recovery codes securely.
- Keep authentication devices updated and protected.
- Report suspicious login attempts immediately.